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Organ-Aware Mixture-of-Experts Framework for Generalized Pan-Tumor Segmentation

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Organ-Aware Mixture-of-Experts Framework for Generalized Pan-Tumor Segmentation

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Abstract

Medical image segmentation aims to delineate the anatomical structures of organs by using different imaging modalities. However, segmentation remains challenging because organs vary anatomically and imaging modalities exhibit distinct feature characteristics. To tackle these complexities, we introduce an Organ-Aware Mixture of Experts (OAMoE) for generalized pan-tumor segmentation, a pioneering segmentation framework that incorporates anatomical prior knowledge within a two-tier expert selection mechanism. We also introduce the concept of expert pooling. The OAMoE categorizes input scans into pools of experts guided by anatomical knowledge, then activates the top two experts in each pool via an adaptive attention-based routing strategy. This layered Mixture-of-Experts (MoE) architecture fosters specialization among the experts while ensuring computational efficiency and interpretability. Our model was evaluated across seven multimodal datasets spanning three anatomical regions: the brain (BraTS2019, BraTS2020, BraTS2021, MSD, OASIS), the breast (US3M), and the ovary (MMOTU). Extensive experiments demonstrate that OAMoE achieves state-of-the-art performance compared to recent large-scale deep models and organ-specific networks. Similarly, the model has demonstrated strong generalization on an unseen dataset, i.e., NSCLC (lung). The average Dice is up to 88.5% on seen datasets, reflecting an accuracy improvement of 2~6%. On unseen datasets, our method improves the accuracy by 1~4%. These results highlight OAMoE's robustness and transferability in cross-organ, cross-modality scenarios. Furthermore, attention visualizations show that the model dynamically allocates tasks to experts sensitive to specific modalities and anatomical features. These findings support future development of interpretable and adaptive clinical AI.

Keywords: Mixture-of-Experts, Medical Image, Tumor Segmentation, Attention

1. Introduction

Medical image segmentation is an essential process in numerous clinical applications, facilitating the precise delineation of anatomical structures and pathologies, which is vital for diagnosis, surgical planning, and treatment [1][2]. In specific scenarios such as radiotherapy planning, it is imperative to simultaneously identify tumors and adjacent organs-at-risk to minimize adverse effects. Hence, accurate multi-organ segmentation is paramount. Nonetheless, this task presents significant challenges due to the heterogeneity inherent in medical data [3][4]. Between patients, organs and lesions vary in shape and size. This often causes under-segmentation of small structures due to class imbalance and scale variability [5]. Consequently, accurately segmenting multiple organs within a single model becomes problematic. Additionally, modalities such as CT, MRI, and PET have distinct characteristics. Datasets from different institutions also exhibit distributional biases [6]. Moreover, low contrast and the presence of noise in scans can obscure the demarcation between adjacent tissues. In datasets characterized by low contrast, noise can introduce artifacts that adversely affect segmentation outcomes. These factors complicate the use of multi-organ segmentation models across diverse datasets. Thereby restricting their generalization capabilities.

A number of newer medical datasets incorporate multimodal information to address the complexity of medical images. Each modality within the same dataset often contains different types of information. Such as in brain tumor datasets, the T1 modality is commonly employed for localizing tumors, the T1ce modality for delineating boundaries, and the Flair modality for assessing tumor type and progression [7]. However, techniques such as early fusion and the use of shared encoders have been implemented to integrate modalities like MRI and CT. Simple fusion approaches frequently fail to fully leverage the complementarities across modalities [8][9][10]. Present-day networks typically fall short in efficiently fusing information from multimodal medical data. Determining how to optimally utilize multimodal information without compromising model efficiency and performance remains a critical challenge in the field of medical multimodal segmentation.

The Mixture of Experts (MoE) architecture represents a sophisticated neural network paradigm that enhances both the expressive capabilities and computational efficiency of models through conditional computation. This architecture comprises multiple parallel sub-networks known as ‘experts’. Each specialized in learning distinct patterns or sub-tasks within specific input feature spaces. The innovative aspect of MoE lies in its utilization of a gating function. This allows for dynamic selection of a few pertinent experts during the inference process based on the input characteristics. This selective activation significantly mitigates computational demands while preserving model performance [11]. The strategy of engaging only the most relevant experts according to the input features promotes specialization and minimizes interference from experts irrelevant to the task at hand [12]. Despite these advantages, current MoE methodologies exhibit notable deficiencies in segmentation tasks. Many MoE segmentation networks employ a rigid routing mechanism where each input triggers only one expert. Thus, failing to leverage the potential synergies from the combined expertise of multiple experts. Conversely, soft gating mechanisms tend to amalgamate outputs from all experts. Thereby diluting the distinct expertise of individual experts. Moreover, most existing studies on MoE segmentation are confined to

certain organs or imaging modalities.

We propose a MoE-driven framework with dual-stage routing. It targets generalization and multimodal information integration in multi-organ segmentation. We developed an anatomical recognition module to classify input scans into designated anatomical regions and direct them to specific expert pools tailored to those regions. Subsequently, a heterogeneous attention gating mechanism within each expert pool will dynamically select the two most pertinent experts for the segmentation task. The contributions of our work are summarized as follows:

1. We introduce an **Organ-Aware Mixture of Experts (OAMoE) framework for medical image segmentation of different tumors**. This framework diverges from conventional shared-encoder architectures by incorporating explicit expert routing and an expert pool mechanism. Such a mechanism dynamically assigns distinct anatomical or modality-specific regions to specialized experts. This architecture addresses the common issues of overfitting and modality and organ interference observed in fully shared networks, thereby substantially enhancing the model's capacity to generalize across diverse clinical datasets.
2. We propose an **Asymmetric Attention Mechanism for dynamic expert selection**. Instead of using static or soft allocation for all experts, we adopt a Top-2 routing mechanism based on query-critical attention score guidance. In this mechanism, expert expertise is judged by how well it captures overall structural information. This allows selective interaction with the most relevant experts within each expert pool. This strategy not only improves segmentation accuracy by exploiting expert expertise, but also improves computational efficiency by avoiding the involvement of irrelevant experts.
3. We propose a **Style-Aware Normalization segmentation module that can specifically link the expert pool via a Top-2 fusion mechanism with normalization**. It is a good solution to the problem of incompatibility between different datasets in the expert pool and the introduction of different sites. We also validate the effectiveness of the method on the tasks of multimodal brain tumor segmentation and breast and ovarian cancer segmentation, as well as perform generalization tests with the introduction of different organ datasets. These results confirm the generalizability, operational efficiency, and robustness of the proposed architecture in generalized medical image segmentation tasks.

2. Related Work

2.1 Mixture-of-Experts for Medical Image Segmentation

Mixture-of-Experts (MoE) decomposes a model into multiple specialized subnetworks ("experts") and a gating module that activates only a subset per input. This design increases capacity while controlling inference cost and reducing interference among sub-tasks [13]. Early hierarchical MoE formalized probabilistic routing and expert competition/cooperation; these ideas underpin modern conditional computation for vision and medical imaging. Surveys further organize algorithms, training-stability techniques, and deployment considerations, and conclude that MoE is well matched to heterogeneous inputs—a defining property of clinical imaging across modalities, scanners, and institutions.

For medical image segmentation, recent work adapts MoE to medical heterogeneity along several axes. First, modality-aware experts dedicate branches to CT/MRI/US, with a gate selecting the most relevant experts during inference to exploit modality-specific cues while preserving adaptability [24]. Second, parameter-efficient expansion uses low-rank factors to introduce experts for new organs or protocols, mitigating catastrophic forgetting in continual settings [25]. Third, personalized/site-aware MoE trains institution-specific experts and fuses them via gating to capture acquisition idiosyncrasies without discarding global knowledge [26]. Finally, patch-level MoE routes anatomical sub-regions to different experts, addressing within-image heterogeneity [27].

Two challenges recur in segmentation. (i) Routing rigidity vs. averaging: hard single-expert activation can miss complementary expertise on mixed-appearance lesions, whereas uniform soft averaging dilutes competencies and increases compute; principled conditional activation is therefore essential. (ii) Narrow expert definitions: experts partitioned by modality or site often excel in-domain yet become dataset-tailored, limiting portability to new organs, protocols, or mixed multi-organ scenes. These gaps motivate MoE designs that retain the benefits of conditional computation while scaling beyond single-modality or single-site partitions, and that remain robust and interpretable under distribution shift. Our framework follows this direction: it leverages expert specialization and conditional activation while directly addressing the limitations above, targeting broad generalization across multi-organ, multi-modality scenarios.

2.2 Attention Mechanisms in Mixture-of-Experts

Attention has been widely adopted in medical image segmentation to enhance saliency and suppress irrelevant context—from attention gates in U-Net decoders to capsule-style iterative routing that preserves part–whole relations [28][29]. Within MoE, attention typically appears in two complementary roles. First, intra-expert attention (channel, spatial, or self-attention) strengthens feature extraction within each expert and improves delineation of small or low-contrast structures. Second, inter-expert attention (attention-like gating) scores the affinity between an input representation and expert embeddings, enabling conditional activation (e.g., top-k selection) without uniformly increasing compute. Together, these mechanisms support specialization and maintain efficiency in heterogeneous clinical data.

Early medical studies have shown that attention gates focus decoding on anatomically relevant regions and improve boundary precision without resorting to complex multi-stage pipelines. Subsequent variants added boundary-aware and multiscale convolutional attention to sharpen organ edges and stabilize multi-resolution cues. These modules are now embedded inside expert backbones to raise per-expert discriminability [30][31]. Meanwhile, MoE models pair attention-enhanced experts with gating networks. Gate scores are commonly computed as dot products between a query (from input features) and expert keys, followed by either sparse activation (hard or top-k) or soft mixing. This pattern appears in modality-aware, parameter-efficient, personalized, and patch-granular MoE designs.

Despite these advances, two limitations persist. First, attention in many medical MoE systems remains primarily intra-expert: it improves feature quality but leaves routing to rigid heuristics

(single-expert hard routing) or uniform soft averaging, which can underuse complementary expertise or blur competencies. Second, most studies couple attention with task- or dataset-specific expert definitions (e.g., modality or site partitions), reducing portability under distribution shifts—new organs, unseen protocols, or multi-organ scenes—even when intra-expert attention is strong. These observations motivate designs in which attention informs both expert specialization and learned, input-adaptive routing, with interpretable scores and sparse conditional computation, so that MoE benefits extend from in-expert feature refinement to end-to-end expert selection in heterogeneous clinical environments.

2.3 Multi-Organ and Multi-Modal Tumor Segmentation

Early deep learning solutions for tumor segmentation were typically organ-specific: models would be trained separately for brain, breast, ovary, etc., often achieving state-of-the-art results when tailored to one context. For example, a standard pipeline like nnU-Net can be configured to each dataset. But requires a distinct model per organ or task [22]. Zhang et al. propose a single model with a task-specific encoding that generates dynamic filter weights. Allowing segmentation of different organ tumors without training multiple networks [14]. Their DoDNet learned to segment liver, kidney, pancreas, and colon tumors (among others) within one architecture. Outperforming separate models by leveraging shared representation while adapting to each organ. Pang et al. introduce CTumorGAN, an end-to-end GAN that learns to segment “any kind of tumor” in CT images [15]. These and related multi-organ segmentation frameworks demonstrate the feasibility of a single model handling diverse tumor sites. However, they typically assume a uniform imaging modality (e.g. only CT in the above cases) and rely on either hard-coded task identifiers or separate output heads for each organ.

In parallel, extensive effort has been devoted to multi-modal tumor segmentation. The goal is to fuse information from multiple imaging modalities to improve lesion delineation. The straightforward strategy is to combine all channels through a shared 3D U-Net or encoder–decoder model [16]. However, simple modal fusion may result in incomplete information. To combat this, researchers have proposed dedicated fusion architectures. For example, HyperDenseNet connected modal streams through dense cross-connections to encourage cross-modality learning in brain tumor MRI segmentation [17]. Swin UNETR and SwinBTS introduce self-attention layers that span across MRI modalities, capturing long-range dependencies and inter-modal relationships to improve brain tumor segmentation [18][19]. These methods report better tumor core and boundary detection by allowing the model to attend to the most informative modality features at each location. Despite such advances, efficiently integrating multi-modal data remains challenging.

Recently, researchers have begun to tackle both axes of diversity—organ and modality—within one framework. A truly general segmentation model must handle different organs (with distinct morphology and pathology) and different imaging modalities or protocol variations [20]. And multimodal segmentation networks are usually confined to one anatomical region [21]. More general-purpose strategies have started to emerge, such as large-scale segmentation models trained on many tasks or prompt-driven frameworks that switch outputs based on an indicated target. However, these solutions often still treat each task independently at inference or lack mechanisms

to prevent negative transfer between tasks [23]. In summary, no prior work fully resolves the conflicts arising from multi-organ, multi-modal learning. Models address multi-organ generalization but assume homogeneous inputs, whereas modality-fusion methods ignore cross-organ variation. This gap motivates the innovations in our framework. By introducing organ-aware expert routing alongside dynamic Top-2 expert cooperation and a style-aware normalization (SANS) module, our approach explicitly tackles the joint heterogeneity.

3. Method

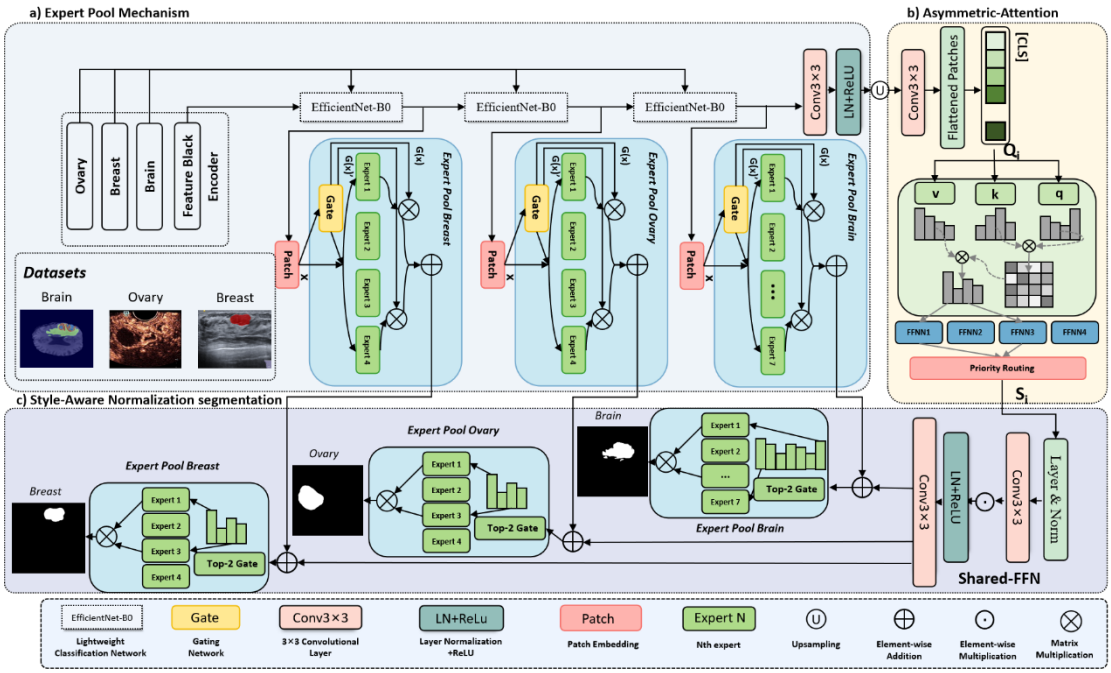

Figure 1. Overall architecture of the proposed OAMoE framework. (a) Expert-pool structure and data-to-pool assignment. (b) The engagement process of the Asymmetric Attention Mechanism. (c) Top-2 routing with Style-Aware Normalization (SANS).

As shown in Figure 1, the OAMoE is proposed for the segmentation of multi-organ tumors. The model employs the concept of an expert pool instead of direct connections between experts in conventional MoE models. The initial stage involves identifying anatomical sites and allocating corresponding expert pools. The anatomical categories of the input images are automatically discriminated by a lightweight EfficientNet-B0 [32] classifier and assigned to the respective pools of expert models for each organ. Then, an asymmetric-attention router selects the Top-2 experts in the pool. We apply SANS to the fused representation before the shared decoder to produce the final segmentation.

3.1 Organ-Aware Mixture-of-Experts Framework

The Organ-Aware Mixture-of-Experts (OAMoE) organizes segmentation capacity into expert pools that are activated conditionally per input. A lightweight anatomical recognizer first assigns each image to the most probable region (e.g., brain, breast, ovary). Computation is then restricted to the corresponding pool. Within that pool, a gating module ranks experts and activates a small cooperative subset to produce the final prediction. In this way, OAMoE scales capacity via

conditional computation while preserving interpretability (organ-/phenotype-specific experts) and efficiency (only a few experts are evaluated). As shown in Fig. 1, the recognizer uses EfficientNet-B0 and precedes the encoder–decoder segmentation backbones; expert activation and fusion yield the output mask.

Each organ/domain maintains a heterogeneous pool whose experts differ in architectural bias and clinical focus. For brain MRI, we include seven specialists: four unimodal experts (T1, T2, T1ce, FLAIR) and three multimodal experts (e.g., T1+T1ce; T2+FLAIR). These designs exploit complementary sequence appearances to improve lesion characterization. For breast and ovarian CT, four experts per organ emphasize distinct phenotypes (e.g., fuzzy-border homogeneous lesions; high-density structural masses; multicystic morphology; an aggregation/all-inputs expert), providing coverage of diverse presentations.

After anatomical assignment, OAMoE performs attention-guided ranking within the selected pool and activates a compact subset for cooperative inference (see Sec. 3.2 for the scoring).In brain MRI, we use dynamic Top-2 activation. The gate selects the two most relevant experts; their masks are fused by learned weights. One expert can refine boundaries while the other stabilizes global structure. In breast/ovary CT, we adopt fixed Top-2 activation. Patterns are more stable, so a consistent complementary pair is used without the overhead of dynamic routing. This hybrid strategy balances adaptability and computational cost across organs. Ablations show that selective cooperation outperforms both single-expert routing and uniform averaging.

The anatomical recognizer (EfficientNet-B0) operates upstream to gate organ-specific branches. Experts adopt established encoder–decoder designs (e.g., nnU-Net variants such as UNet++ [42] and ResUNet++ [50]) and are trained within their pools. The routing module is inserted before decoding to select the pool via anatomy and activate the subset via attention. This yields hierarchical, non-interfering decisions. The configuration is modular (experts can be added per organ) and portable (experts remain self-contained), and it is compatible with normalization-based robustness modules.

OAMoE combines anatomy-aware pool selection with subset-wise expert cooperation to deliver reliable segmentation across heterogeneous clinical inputs. By constraining the search space with anatomical priors and concentrating computation on a few complementary specialists, the architecture retains MoE's capacity benefits while controlling inference cost and preserving interpretability.

3.2 Asymmetric Attention Mechanism

To facilitate the automatic selection of expert models, we have introduced an Asymmetric Attention (Asy) mechanism within the expert pool. This mechanism shall facilitate the dynamic allocation of the most appropriate expert for a given input.

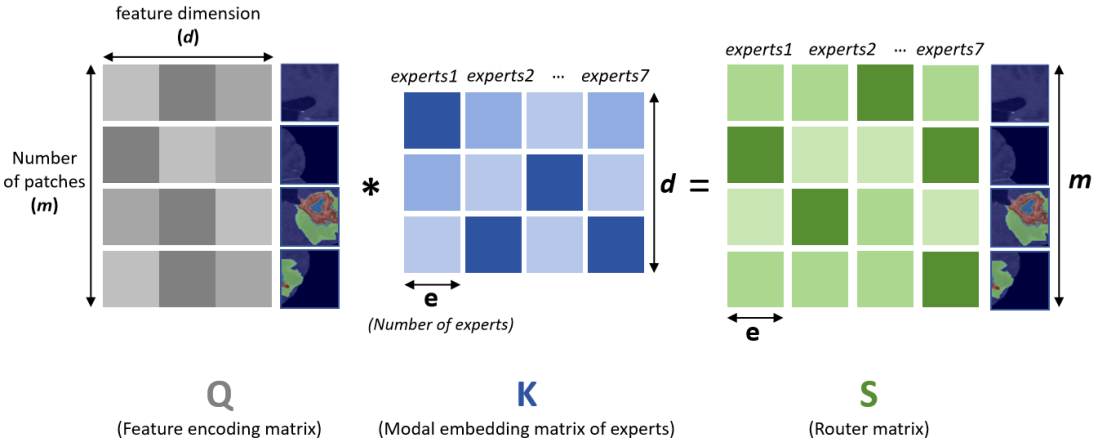

Figure 2. Schematic of the asymmetric attention mechanism. The router represents the input image features that have been transformed into query vectors (Query) by the encoder.

Each expert has an embedding 'Key'. The router compares the 'Query' with each Key and routes by similarity. Specifically, consider the input feature vector as q and the embedding vector of the i-th expert as K_i. The degree of alignment between q and K_i is quantified through the vector dot product, and the routing score for the expert is computed as follows:

$$S_i = q \cdot k_i \tag{1}$$

Here, a higher value of S_i suggests a greater alignment between the input features and the feature pattern of the ith expert's model. The similarity scores for all experts are compiled into a vector $S = [S_1, S_2, ..., S_N]$, where N represents the number of experts in the pool. The router then implements Softmax normalization on the score vector S to derive the normalized weight α_i for each expert:

$$\alpha_i = \frac{exp(s_i)}{\sum_{j=1}^{N} exp(s_j)} \tag{2}$$

Equation (2) ensures that all weights are non-negative and their sum equals 1, thereby prioritizing the experts with higher scores through larger α_i values. Subsequently, the router selects the two highest-scoring expert models from the pool to engage in the current tumor segmentation inference, based on the magnitude of their attentional weights. The expert associated with the largest weight is designated as Top-1 (with weight α_i and expert index i), and the expert with the second largest weight as Top-2 (with weight α_j and index j). Notably, for this particular input, only these two expert models are activated for the segmentation prediction. We fuse the two predictions by a weighted sum:

$$M_{fused} = \alpha_{i^*} M_{i^*} + \alpha_{j^*} M_{j^*} \tag{3}$$

In the equation above, M_i and M_j denote the tumor probability maps or segmentation masks produced by the Top-1 and Top-2 experts, respectively. The proposed weighted fusion strategy integrates the distinctive predictive capacities of these experts to yield a more accurate final

segmentation result. The novel heterogeneous attention routing mechanism adaptively activates different experts based on the input features, thus facilitating a unidirectional selection predicated on feature similarity. Intuitively, the optimal expert is aligned with the image features, rather than being pre-assigned according to a predetermined rule.

In the context of brain MRI analysis, the described asymmetric attention routing mechanism effectively determines the expert best suited to the modal features of the current case. For instance, in scenarios where T2-weighted information is critical for accurate segmentation, the mechanism preferentially selects the expert specializing in T2 features, thereby enhancing the segmentation outcome.

Conversely, within the expert pools for breast and ovarian CT scans, a more streamlined expert selection process is implemented where tumor image features exhibit relatively stable and explicit variations. This process involves the consistent activation of the two most effective and complementary experts (Top-2) within each CT expert pool during the inference phase. Despite each pool includes four experts, only the two most proficient are engaged in the actual segmentation process. The fixed dual-expert approach balances accuracy and complexity in CT: it integrates two complementary experts and avoids the overhead of dynamic routing.

3.3 Style-Aware Normalization Segmentation (SANS)

We introduce Style-Aware Normalization for Segmentation (SANS) as a lightweight, plug-in normalization module that operates after attention-guided expert selection and Top-2 fusion and before the shared decoder. Unlike BN/IN in the backbones, SANS standardizes the fused features from the activated experts. It suppresses sample-specific style while preserving anatomical content, and it adds no extra losses. SANS is independent of routing and expert selection. Consider a feature map for one input image at the channel c, $F_c \in R^{H\times W}$. SANS computes instance-wise statistics.

$$\mu_c = \frac{1}{HW}\sum_{h,w} F_c(h,w), \;\; \sigma^2 = \frac{1}{HW}\sum\nolimits_{h,w} (F_c(h,w) - \mu_c)^2 \tag{4}$$

In Eq. (4), μ_c and σ_c^2 are per-sample, per-channel statistics over spatial (h,w). They do not depend on batch statistics (unlike BatchNorm) and are closer to InstanceNorm. The goal of InstanceNorm is to remove the 'style' drift caused by differences in intensity scale, contrast, and texture at the sample level. Since channels typically correspond to a class of detectors with semantic or anatomical substructures. Spatially solving for statistics by channel is equivalent to establishing an adaptive intensity baseline and dynamic range for each detector. Subtracting μ_c removes global luminance and offset differences; normalizing by σ_c^2 suppresses contrast/gain differences. This unifies activations across protocols, stabilizes the decoder's input, and improves cross-domain generalization.

If the response styles of two experts for the same channel differ significantly (e.g., one expert responds more strongly under high-contrast scanning and the other is more stable under

low-contrast scanning), the covariance term and the weighted variance term cause the fusion distribution to appear to be on an inconsistent scale. The channel-by-channel, sample-by-sample normalization of F in Equation (4) is equivalent to unifying this 'cross-expert style mismatch' into a zero-mean, unit-variance common domain, which is then given to the decoder - this is where we put SANS. This is the core reason why we put SANS 'after attentional selection and Top-2 fusion': it acts directly on the cooperative representation, without destroying expert complementarity, and explicitly cancels out cross-domain and cross-expert style noise.

SANS applies per-channel normalization followed by a learned affine transform:

$$\hat{F}_c(h,w) = \gamma_c \frac{F_c(h,w) - \mu_c}{\sqrt{\sigma_c^2 + \varepsilon}} + \beta_c \tag{5}$$

Here γ_c and β_c are trainable scale and bias. Equation (4) filters global appearance (e.g., brightness, contrast, coarse texture) on a per-sample basis, while γ_c, β_c recover discriminative scaling needed for anatomy and lesion cues.

By acting after expert cooperation, SANS preserves complementary expert competencies while neutralizing cross-domain style, and is fully compatible with our hierarchical routing design. It works with unimodal experts (e.g., MRI sequences) and multimodal experts alike. It also coexists with the OAMoE routing: any expert selected by the gate benefits from the same style-robust features. SANS facilitates transfer to unseen anatomical regions by standardizing low-level appearance statistics that vary across organs, without assuming organ-specific normalization. Consequently, experts retain content-focused features when encountering previously unseen organs. Complementing the routing stage with domain-robust normalization. By standardizing feature style at the source. SANS reduces the distribution shift between training and deployment domains. Experts focus on anatomical content rather than scanner or protocol idiosyncrasies. This leads to more stable features and better segmentation on unseen datasets, without extra supervision or computational overhead.

3.4 Network Architecture and Training Strategies

The initial phase of the EfficientNet-B0 target recognizer is treated as an autonomous predecessor module, which is trained concurrently with the backbone segmentation network, nn-UNet. The categorization output of this module is employed to control the activation of subsequent segmentation branches. The backbone segmentation network adopts an encoder-decoder architecture. Leveraging multi-scale features during the encoding stage and applying a progressive up-sampling method in the decoding phase to aid in the accurate reconstruction of the segmentation. Before the commencement of the decoding process, an expert pool routing module is integrated. This procedure begins with the selection of the appropriate expert pool based on the identified anatomical category, followed by the deployment of a heterogeneous attention router to align and activate the expert model within the chosen pool. This modular configuration guarantees that the identification of anatomical sites and the routing of expert selection operate sequentially as two independent network components, thus enabling hierarchical and non-interfering routing decisions.

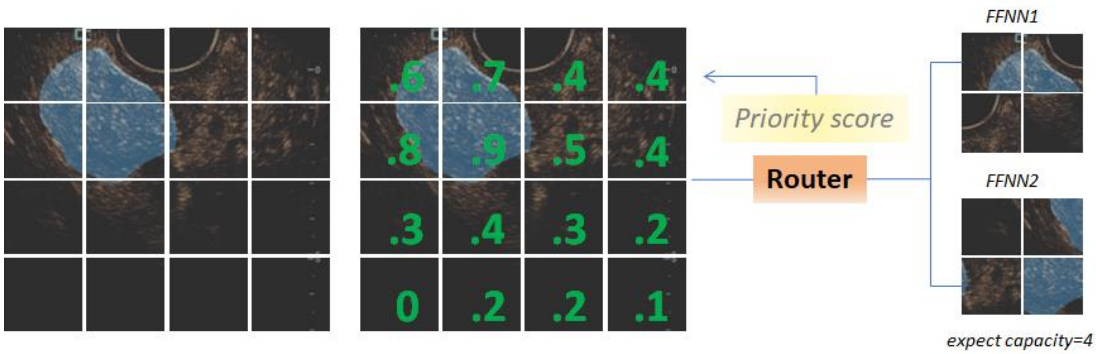

Figure 3. Targeted regional priority mechanisms. The ovarian cancer image data serves as an example, where the green numerals indicate weights; the proximity to the Mask correlates with a higher relevance score.

Simultaneously, as depicted in Figure 3, the network adopts a routing area priority mechanism, where it prioritizes discriminative routing for anatomical sites before executing expert selection in the respective sub-network. This method promotes a phased decision-making process, enhancing both the routing precision and the overall efficiency of the network.

During model training, a multi-task joint optimization strategy is utilized to train the classifier, router, and segmentation expert in an end-to-end manner. The training loss incorporates a combination of classification loss for anatomical site identification and segmentation loss for tumor segmentation. The classification loss employs a cross-entropy loss, while the segmentation loss comprises a synthesis of the Dice coefficient and cross-entropy loss to optimize both regional and boundary accuracies. Specifically, the cross-entropy loss for the classification task is defined as follows:

$$L_{cls} = -\sum_{c} y_c \ln p_c \tag{6}$$

where the indicator variable y_c denotes the true category c (where $y_c = 1$ when the sample belongs to the category c, and $y_c = 0$ otherwise). The predictive probability of the model for the category c is represented by P_c.

The segmentation task's loss function comprises two primary components: Dice loss and pixel-level cross-entropy loss. The Dice coefficient, a metric for quantifying the overlap between the predicted and actual masks, is defined as follows:

$$D = \frac{2\sum_x P(x)G(x)}{\sum_x P(x) + \sum_x G(x)} \tag{7}$$

Here, $P(x)$ represents the tumor probability predicted by the model, and $G(x)$ denotes the actual tumor label at pixel x (where 1 indicates a tumor and 0 indicates background). The Dice loss, L_{dice}, is subsequently calculated as $L_{dice} = 1 - D$, with smaller values indicating more precise segmentation outcomes.

Conversely, the pixel-level cross-entropy loss, L_{seg-ce}, is determined by averaging the cross-entropy across all pixel predictions. The total segmentation loss is then articulated as the sum of these two losses:

$$L_{seg} = L_{dice} + L_{seg-ce} \tag{8}$$

The concurrent optimization of L_{dice} and L_{seg-ce}facilitates a balance between the completeness of the segmentation region and the precision of boundary delineation. The introduction of an expert routing mechanism leads to the adoption of a 'soft fusion' approach during the initial training phase. This method employs the expert weights derived from the router to weight and amalgamate the features or prediction outcomes from multiple experts. Subsequently, the gradient is utilized to simultaneously update the parameters of various expert models. In essence, the model's prediction output is represented as the weighted sum of the results from all experts:

$$S = \sum_{i=1}^{N} \alpha_i S_i \tag{9}$$

In this equation, S_i denotes the segmentation result generated by an expert, while α_i represents the attention weight assigned to this expert by the router. This weighted fusion approach ensures that the error gradient is distributed to each expert's parameters in accordance with their respective weight α_i. This methodology aims to facilitate synchronous learning among multiple experts. As training progresses, the strategy shifts to 'hard-routing', wherein the router selects only the most heavily weighted expert (Top-1) to generate segmentation results and conduct error backpropagation. This approach effectively activates the most proficient expert for each training sample, whose output is then utilized solely for loss computation and parameter updates. The training strategy evolves from soft to hard, promoting collaborative learning among experts while bolstering the independent decision-making capabilities of each expert model in later stages. During the inference phase, a hard routing strategy is implemented to expedite the inference process.

For each new instance, the expert model possessing the greatest final weight is deployed for segmentation prediction. Research in brain MRI tasks has confirmed that selecting a fusion of the top two experts can marginally enhance accuracy. Additionally, a sparse routing regularity term, L_{sparse}, is incorporated into the loss function to foster sparsity in routing decisions. This term is designed to restrict the router to activating a minimal number of experts simultaneously:

$$L_{sparse} = \sum_{i=1}^{N} \alpha_i (1 - \alpha_i) \tag{10}$$

L_{sparse} assumes a minimal value when only one α_i approaches 1 while the remaining weights tend toward 0. In contrast, if multiple experts are assigned substantial weights concurrently, L_{sparse} escalates and imposes a penalty on the router during backpropagation.

In summary, the total loss incurred during the training phase can be articulated as:

$$L = L_{cls} + L_{seg} + \lambda_s L_{sparse} \tag{11}$$

Here, the coefficient λ_s functions as the regulatory mechanism that moderates the influence of the sparse regularity term.

The optimization of the aforementioned losses is facilitated through a synergistic process that entails the simultaneous updating of the parameters of individual sub-models and the weights of the router. This cooperative optimization strategy empowers the expert network to autonomously determine the most appropriate amalgamation of experts, predicated on the inputs received. As a result, this approach not only boosts the model's generalization capability and inference efficiency but also ensures the preservation of high segmentation accuracy.

4. Experiments and Results

All experiments were conducted on workstations equipped with six NVIDIA A40 GPUs, each with a capacity of 48 GB. The expert pools within the OAMoE framework underwent a training period of approximately 40 hours, utilizing image inputs of 512 × 512 resolution. Throughout the training and testing phases, all lesion subtypes were amalgamated into a single foreground class, without further subclassification. For the dataset of brain tumors composed of 3D MRI voxels, our analysis was confined to axial slices numbered 70-120, where the tumor region predominantly manifests. Slices outside this range were omitted to circumvent the inclusion of empty background slices.

4.1 Datasets

To comprehensively evaluate the proposed segmentation model, multiple medical image datasets encompassing three anatomical regions—brain, breast, and ovary— were included. An overview of the datasets is illustrated in Figure 4. These datasets cover varied imaging modalities and tumor characteristics.

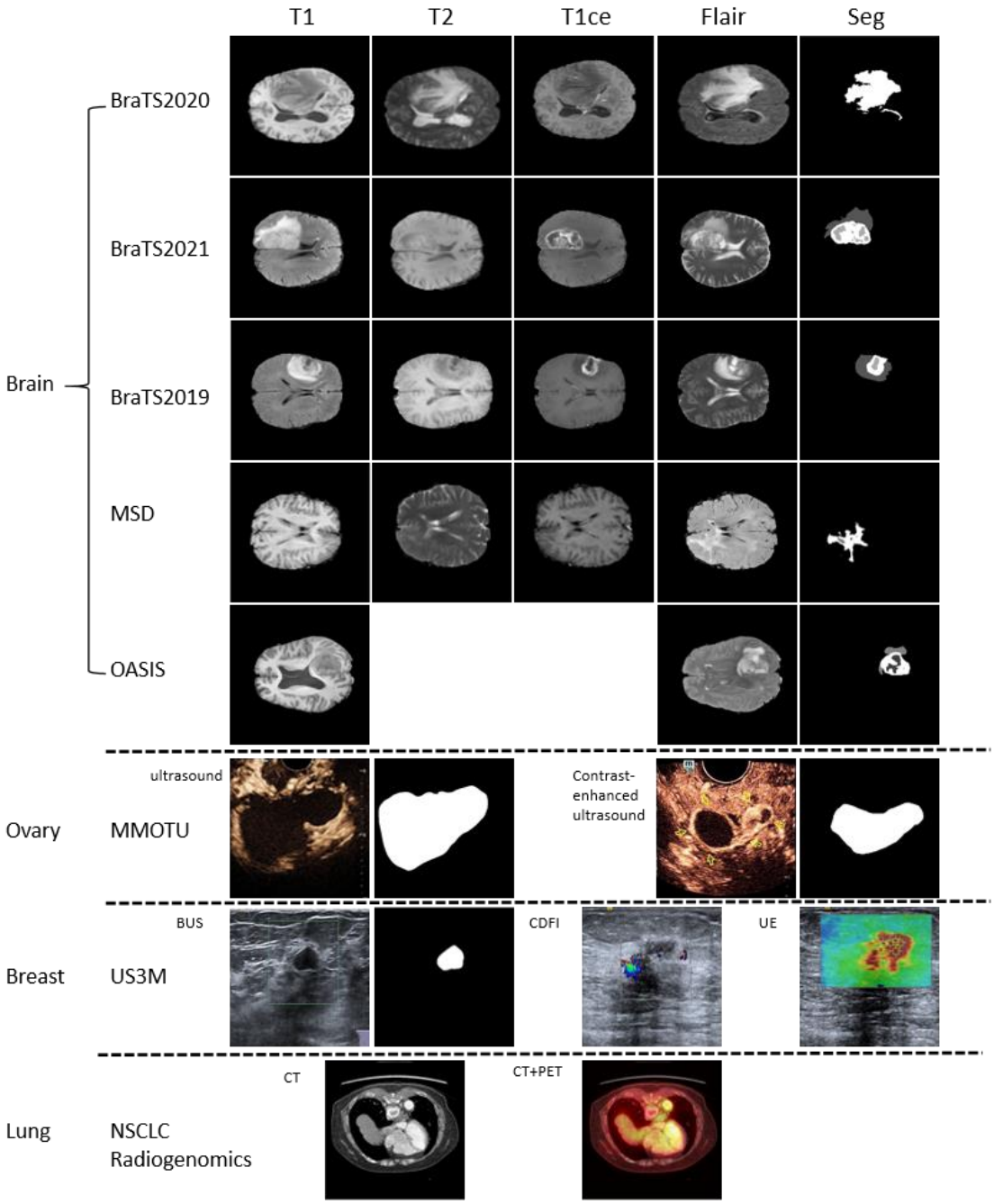

Figure 4. An overview of the types of datasets used in our study. A total of eight multimodal datasets of different organs and modalities were utilized where the lung dataset is used for our ablation studies in asymmetric attention.

Segmentation of Brain Tumors: The datasets utilized included the BraTS2020 dataset [33], the Brain Tumor Segmentation task from the Medical Segmentation Decathlon (MSD) [34], and the OASIS brain MRI dataset [35]. All BraTS and MSD datasets provide multimodal MRI sequences (T1, T1ce, T2, FLAIR) and have been manually annotated by expert radiologists to delineate tumor subregions (enhancing tumor, tumor core, whole tumor). For the OASIS dataset, we only have T1 and FLAIR modalities. For the purposes of this experiment, these subregion labels were consolidated into a unified tumor category. To evaluate the model's generalization capabilities,

additional assessments were performed on the previously unseen BraTS2021 [36] and BraTS2019 [37] test sets, which had not been utilized during the training phase.

Segmentation of Breast Lesions: The US3M multimodal breast ultrasound dataset [38] was employed, comprising 1,532 images from 248 patients across BUS, CDFI, and UE; images include radiologist-provided pixel-level lesion masks and benign/malignant labels.

Segmentation of Ovarian Tumors: For the pelvic region, the Multi-Modality Ovarian Tumor Ultrasound (MMOTU) dataset [39] was utilized. This publicly available dataset features greyscale 2D ultrasound and color Doppler (contrast-enhanced) ultrasound images sourced from multiple hospitals, encompassing over 1,100 patient cases. Radiologists provided pixel-level segmentation and boundary annotations for ovarian masses.

Segmentation of Lung Tumors: We used the NSCLC-Radiogenomics dataset [68] for Lung Tumors test. The NSCLC-Radiogenomics dataset comprises 211 NSCLC subjects with paired thoracic CT and PET/CT. Specifically, the dataset provides CT tumor segmentation maps and semantic annotations, as well as PET quantitative values, genomic profiles (mutations, microarrays, RNA-seq) and clinical outcomes (including survival).

4.2 Evaluation Metrics

Five standard metrics were employed to assess performance quantitatively:

$$DICE = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{12}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{13}$$

$$ACC = \frac{TP + TN}{TP + FP + FN + TN} \tag{14}$$

$$Recall = \frac{TP}{TP + FN} \tag{15}$$

$$\mathrm{d_H}(\mathrm{X},\mathrm{Y}) = \max\{\mathrm{d_{XY}}, \mathrm{d_{YX}}\} = \max\{\max_{\mathrm{x}\in\mathrm{X}} \min_{\mathrm{y}\in\mathrm{Y}} \mathrm{d}(\mathrm{x},\mathrm{y}), \max_{\mathrm{y}\in\mathrm{Y}} \min_{\mathrm{x}\in\mathrm{X}} \mathrm{d}(\mathrm{x},\mathrm{y})\} \tag{16}$$

Within these equations, the terms TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. The metrics Dice and IoU evaluate the extent of overlap between the predicted segmentation and the actual lesion area. Accuracy measures the overall pixel-level correctness, while recall reflects the proportion of true lesion pixels accurately identified, known colloquially as the true positive rate. HD95, indicative of the 95th percentile of the Hausdorff distance, measures the distance between the segmentation result and the actual lesion boundary. The metric is particularly robust as it calculates the maximum error distance while excluding the most significant 5% of outliers [40]. Lower HD95 values suggest a more precise alignment of tumor contours. Collectively, these metrics gauge segmentation quality, encompassing region overlap, lesion coverage, and boundary precision.

4.3 Performance Comparison of State-of-the-Art for Generalized Medical Image Segmentation

Table 1. Comparison of our model and the state-of-the-art by using different multimodal datasets.

Model&Dataset	Brats2020	MSD	OASIS	Brats2021	Brats2019	US3M	MMOTU
M^4oE[24]	86.21	86.79	87.23	83.04	79.32	88.47	**88.21**
PG-SAM[55]	85.34	83.69	85.21	80.78	77.63	84.09	81.29
EICSeg[56]	83.24	81.25	79.38	77.93	79.21	81.58	80.23
MedSAM[67]	87.61	88.12	**88.37**	85.14	80.58	87.61	83.24
UniverSeg[57]	82.77	88.69	85.34	81.96	77.37	83.55	81.29
TransUnet[46]	73.24	78.31	83.65	83.39	79.82	81.36	79.35
UNet[41]	80.01	83.30	81.29	81.47	79.85	85.21	79.21
MoME[44]	81.69	82.12	81.31	82.54	79.38	79.11	84.14
nnU-Net[22]	78.03	77.85	77.64	81.79	79.19	85.33	77.81
OAMoE (Ours)	**90.83**	**89.08**	87.74	**86.39**	**85.61**	**89.93**	87.91

As shown in Table 1, OAMoE achieves higher Dice and IoU scores and lower HD95 values than general-purpose baselines, including nnU-Net [22], MedSAM [67], and the MoE variant M4oE [24], across four evaluated organs. The improvement is consistent in in-distribution validation and becomes more evident in out-of-distribution scenarios, such as BraTS2021 for brain MRI and the ultrasound benchmarks with large appearance variations. Findings from Tables 2–4 further indicate that OAMoE sustains comparable or better recall while offering clear advantages in boundary accuracy, reflected by the lowest HD95 on each organ. Thes hybrid Dense-Transformer e findings suggest that conditional specialization enhances robustness under protocol and site shifts, and that expert cooperation provides advantages over single-backbone CNN, Transformer, or diffusion-based models.

Three architectural choices account for the observed results. First, anatomy-aware pooling confines representation learning to organ-specific subspaces, reducing interference across modalities and scanners and thereby supporting stable expert specialization and generalization. Second, Top-2 cooperative routing selects complementary experts at uncertain boundaries; this avoids the errors of Top-1 selection and the over-smoothing of uniform averaging, leading to consistent gains in Dice and HD95 without adding significant computational cost. Third, feature refinements contribute at different levels: SANS reduces style variation at the instance level, GCSA improves boundary delineation across scales, and RSF encourages structural coherence in cases with high variability or limited samples. These components collectively enhance robustness on unseen datasets and novel anatomical conditions. The improvements reported in Tables 2–4 and the error reductions illustrated in Figure 5 align with the intended effects of pooling, routing, and normalization within the proposed architecture.

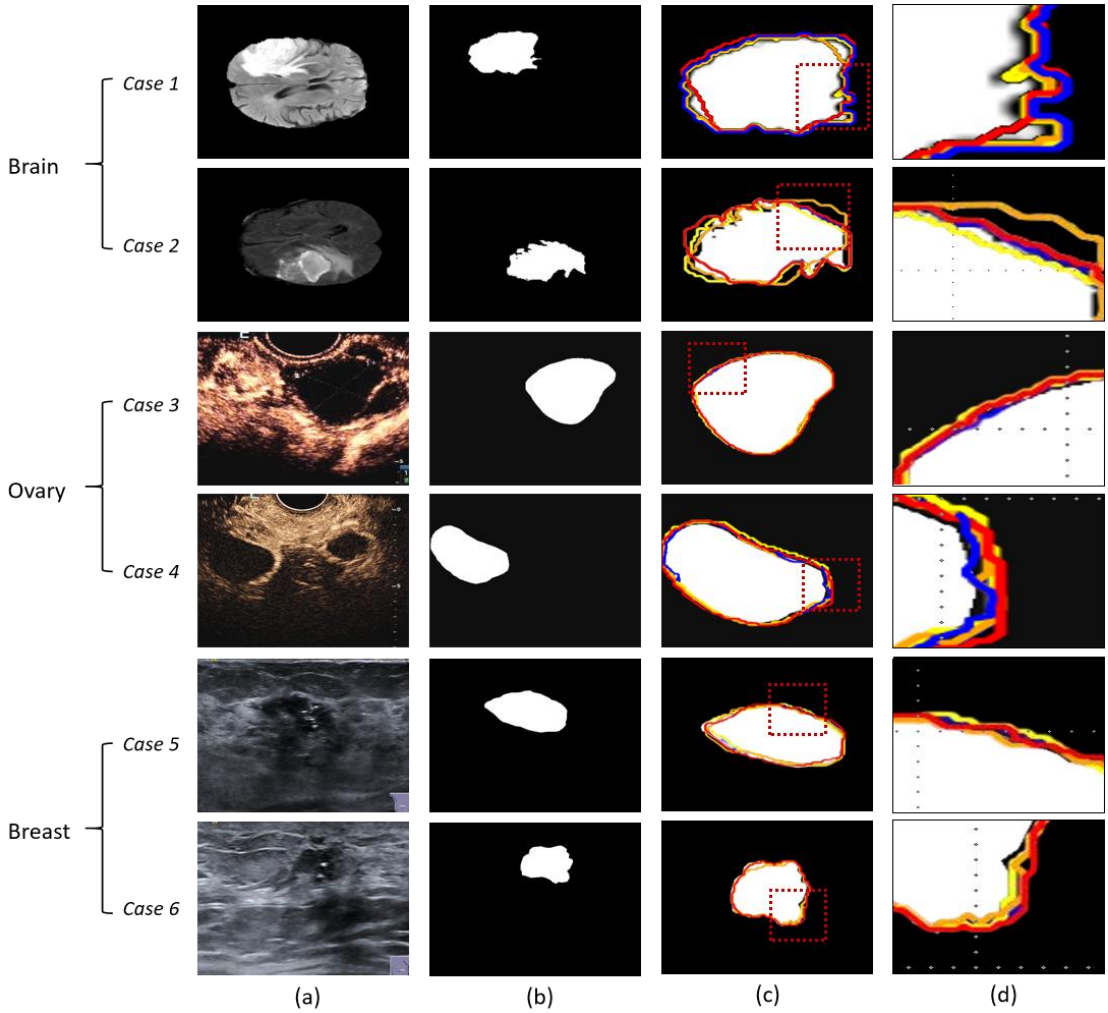

Figure 5. Comparison of different model segmentation results for brain (case 1, 2), ovary (case 4, 5) and breast (case 5, 6). (a) Original image input. (b) Their corresponding ground truth (GT). (c) Segmentation result of different models acting in GT. (d) Specific details of segmentation where the blue line is OAMoE, the yellow is MedSegDiffv2, the orange is ViT, and the red is MoME.

As illustrated in case 2 and case 4 of Figure 5, ViT [45] frequently over-segments along the enhancing rim and produces false positives in peritumoral edema. In case 1 and case 4, MoME [44] tends to under-segment non-enhancing regions and miss small issues. while MedSegDiffv2 [43] under-segments thin rims and yields unstable boundaries. In contrast, OAMoE provides contours that are closer to the reference masks and reduces both over- and under-segmentation. These improvements reflect the use of anatomy-aware expert pools and Top-2 cooperative routing, which combine complementary cues from T1ce and FLAIR. SANS mitigates site-dependent intensity shifts, and GCSA enhances boundary delineation across scales. OAMoE produces contours closer to radiologist annotations and lowers boundary error. Expert cooperation helps capture both low-contrast texture and sharp structural edges, while GCSA suppresses background artifacts. RSF reinforces cross-case structural coherence, and SANS improves robustness across probe types and scanning presets.

The qualitative outcomes that corroborate the quantitative data presented in Table 1. Accordingly,

the OAMoE has demonstrated superior accuracy and completeness in segmentation results, which suggest the exceptional adaptation to diverse visual features across different modalities. This visual enhancement is particularly notable in challenging scenarios characterized by low contrast or complex boundaries, suggesting that certain experts within the OAMoE model pool are adept at identifying specific features that are not effectively captured by generic or single-expert models.

4.4 Performance Comparison of Single Organ Tumor Segmentation

The OAMoE model is compared against each organ-specific advanced model on the corresponding organ-specific benchmark datasets (Tables 2-4).

Table 2. Comparative analysis of brain tumor segmentation

Model	Dice	IoU	Acc	Recall	HD95
MedSegDiffv2[43]	88.26	83.48	97.29	80.22	14.83
MoME[44]	86.74	82.37	99.07	79.89	12.51
ViT[45]	83.55	79.71	98.56	**81.42**	14.15
MedNeXt[58]	87.92	83.62	99.05	81.05	9.98
GBT-SAM[59]	88.04	83.90	99.02	81.00	9.45
OAMoE (Ours)	**88.51**	**84.05**	**99.09**	80.50	**7.53**

Table 2 presents a comprehensive evaluation of brain tumor segmentation on BraTS2020, MSD, OASIS, and BraTS2021. The compared models include MedSegDiffv2 [43], MoME [44], ViT [45], MedNeXt [58], and GBT-SAM [59]. OAMoE attains the best Dice (88.51%) and IoU (84.05%), together with the lowest HD95 (7.53 mm). The gain in HD95 is substantial: compared with MoME [44], the boundary error drops from 12.51 mm to 7.53 mm. This improvement reflects the combined effect of anatomy-aware expert pools and selective cooperation, which stabilize edges under multi-sequence inputs. We note that OAMoE's Recall is lower than ViT [45]. This is consistent with a more conservative decision boundary induced by expert cooperation, which emphasizes precision and reduces false positives. Even with this trade-off, OAMoE provides the strongest balance of accuracy and boundary localization across four datasets. The results indicate that a mixture-of-experts formulation can rival or surpass single-backbone CNN/Transformer and diffusion models when handling heterogeneous MRI appearances.

Table 3. Comparative analysis of breast tumor segmentation

Model	Dice	IoU	Acc	Recall	HD95
FABRF-Net [48]	89.12	87.65	93.01	83.26	14.04
DA-UNet[60]	86.70	85.10	97.90	82.20	12.10
PDPNet[61]	88.20	87.10	98.10	84.50	10.20
HAA-DSUNet[62]	88.75	88.05	98.26	84.90	9.80
MMF-U-Net [63]	88.60	87.90	98.20	84.70	9.95
OAMoE (Ours)	**89.93**	**90.91**	**98.82**	**85.45**	**6.44**

Table 3 reports results on the US3M breast ultrasound dataset, comparing FABRF-Net [48], DA-UNet [60], PDPNet [61], HAA-DSUNet [62], and HTU-Net [63]. OAMoE achieves the highest Dice (89.93%) and IoU (90.91%), and the lowest HD95 (6.44 mm). Ultrasound images

suffer from speckle noise and blurred lesion margins; nonetheless, the anatomy-aware experts capture lesion extent reliably while the cooperative fusion preserves sharp boundaries. Compared with strong CNN/Transformer baselines, OAMoE improves recall without sacrificing contour quality, suggesting better coverage of low-contrast or heterogeneous masses. The overall picture is consistent with our architectural design: conditional specialization controls computation while maintaining robustness to ultrasound appearance variations. These observations confirm that a multi-expert approach is competitive even against methods tailored specifically for breast lesions.

Table 4. Comparative analysis of ovary tumor segmentation

Model	Dice	IoU	Acc	Recall	HD95
VGG-16[47]	86.68	84.51	91.29	**84.55**	14.55
SovaSeg-Net[49]	**87.93**	85.42	93.39	79.33	10.68
SMoFFI-SegFormer [64]	86.90	85.30	94.00	81.20	11.50
MBA-Net[65]	87.60	86.90	94.50	81.70	9.90
Res-ECA-Unet++[66]	87.40	87.10	94.70	81.60	9.20
OAMoE (Ours)	87.91	**88.63**	**95.04**	82.02	**7.84**

As illustrated in Table 4, the performance of the OAMoE model on the Multimodal Ovarian Tumor Ultrasound (MMOTU) dataset is compared against previous ovarian segmentation models such as VGG-16 [47], SovaSeg-Net [49], SMoFFI-SegFormer [64], MBA-Net [65], and Res-ECA-UNet++ [66]. OAMoE attains Dice = 87.91% and IoU = 88.63%, with a low HD95 = 7.84 mm. Ovarian lesions often appear with irregular shapes, multilocular cysts, and fuzzy borders; these factors challenge boundary stability in ultrasound. The expert-pool design addresses this variability by assigning complementary specialists to structural and texture cues, while selective cooperation mitigates local ambiguities. As a result, OAMoE reaches higher recall than most baselines and maintains accurate contours, yielding the best or near-best scores across metrics. The findings support the generality of our approach: a unified MoE framework can handle diverse ultrasound phenotypes without task-specific heuristics, and it scales favorably when datasets exhibit heterogeneous imaging conditions.

Tables 2, 3 and 4 underscore the consistent superiority of OAMoE over numerous organ-specific methods, thereby substantiating the efficacy of the proposed approach. The integration of domain-specific expert knowledge within a unified framework enables OAMoE to achieve both generalizability and state-of-the-art accuracy across various independent datasets.

4.5 Expert Heatmap Visualization

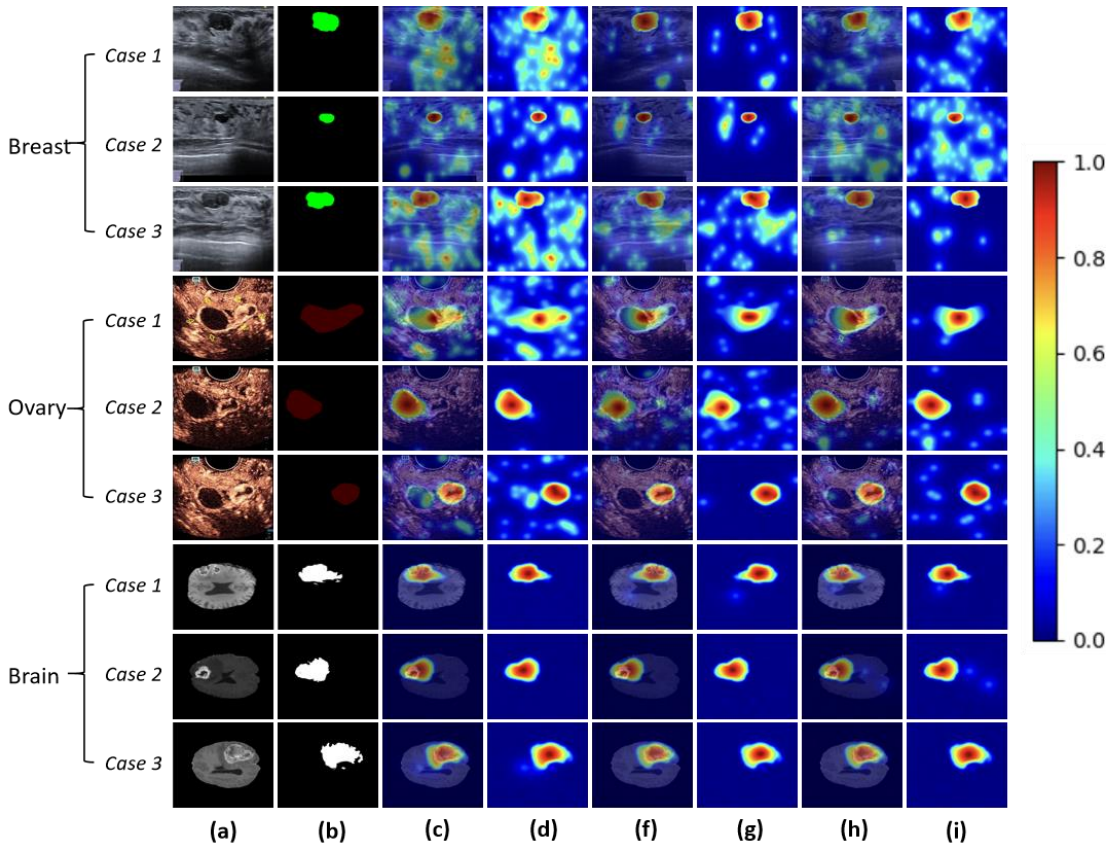

Figure 6. Attention heatmaps from three organ-specific experts. **(a)** Input image; **(b)** Ground truth (GT); **(c)** Attention heatmap A on the input image; **(d)** Attention heatmap prediction output from Expert A; **(f)** Attention heatmap B on the input image; **(g)** Attention heatmap prediction output from Expert B; **(h)** Attention heatmap C on the input image; **(i)** Attention heatmap prediction output from Expert C.

As demonstrated in Figure 6, Expert A, Expert B, and Expert C represent different specialists within distinct organ-specific expert pools. For example, the brain tumor expert pool would consist of Expert A_{brain}, Expert B_{brain}, Expert C_{brain}, and so forth. Each expert develops a unique focal point of attention and demonstrates anatomical proficiency that is contingent on the input modality and dataset characteristics. For instance, in the brain MRI segmentation task, Expert A exhibits substantial focus on Case 1, while Expert C distributes attention more evenly, correlating closely with the original mask in Case 3.

Regarding the CT imaging data within the expert pool, task allocation is exemplified as follows: In breast ultrasound, Expert A is dedicated to delineating the boundaries and shapes of the lesion with heat maps in case 2, Expert B focuses on the internal texture of the lesion in case 3, and Expert C is responsible for delineating the entire background with masks in case 1. Similar configurations were also observed in ovarian ultrasound examinations. Expert A focused on the ovarian mass area in case 1, Expert B focused on the adjacent ovarian boundary and posterior acoustic features in case 3, and Expert C focused on the broader image area in case 2.

These heatmaps show that experts specialize in specific features and sub-regions, as intended. The gate routes inputs to the most sensitive experts, producing a divide-and-conquer effect that improves final segmentation.

4.6 Ablation Studies in Asymmetric Attention

We conduct ablations to quantify each component. The following four datasets were used: brain—BraTS2020, breast—US3M, ovary—MMOTU, and lung—NSCLC. Enabled by the modular design, a lung-specific expert pool can be added without altering other components; at inference, the router activates these experts on thoracic CT, enabling lung tumor segmentation with minimal overhead. Table 5 shows model variants according to organ-aware pooling, the attention router, and sparsity regularization:

A (Full OAMoE): Anatomy-aware pooling on; attention router uses Top-2; sparsity regularization on. This is the reference.

B (Random Pool Assignment): Disable anatomy priors and randomly assign cases to organ pools; router Top-2; sparsity on.

C (Single Generic Pool): Merge all experts into a single shared pool (no anatomy split); router Top-2; sparsity on.

D (Average Fusion): Anatomy pooling on; router uses uniform averaging of all experts in the selected pool.

E (Fixed Top-2 Experts): Anatomy pooling on; activate a fixed complementary pair per organ; sparsity on.

F (Top-1 Expert Only): Anatomy pooling on; router picks the best expert.

G (w/o Sparse Regularization): Anatomy pooling on; router Top-2; sparsity off.

H (Generic MoE, Global Pool): No anatomy split; treat all experts as one global pool with Top-2 routing.

I (Shared Encoder–Decoder): Non-MoE baseline (single backbone; no router; no pools).

J (w/o SANS, Sec. 3.3): Replace SANS with vanilla normalization; other settings follow A.

Table 5. Ablations of OAMoE components: organ-aware pooling, attention router (Top-1/Top-2/Avg/Fixed), and sparsity regularization.

Model variant	Anat. Pool	Attn. Router	Sparse Reg.	Brain (Dice / HD95)	Breast (Dice / HD95)	Ovary (Dice / HD95)	Lung (Dice / HD95)
A. Full OAMoE (Ours)	✓	Top-2	✓	**88.51 / 7.53**	**89.93 / 6.44**	**87.91 / 7.84**	**67.60 / 23.52**
B. w/ Random Pool Assignment	✗	Top-2	✓	86.02 / 9.21	87.35 / 8.67	85.98 / 9.55	65.70 / 25.92
C. w/ Single Generic Pool	✗	Top-2	✓	85.78 / 9.35	86.91 / 8.92	85.47 / 9.87	65.55 / 26.26
D. w/ Average Fusion	✓	Average	✓	87.21 / 8.34	88.04 / 7.89	86.52 / 8.76	66.20 / 25.45
E. w/ Fixed Top-2 Experts	✓	Fixed	✓	87.65 / 8.05	88.97 / 7.12	86.89 / 8.33	67.60 / 24.71
F. w/ Top-1 Expert Only	✓	Top-1	✓	87.93 / 7.98	89.12 / 6.89	87.05 / 8.15	67.20 / 24.88

G. w/o Sparse Regularization	✓	Top-2	✗	87.89 / 8.01	89.25 / 6.71	87.11 / 8.09	66.90 / 25.13
H. Generic MoE (Global Pool)	✗	Top-2	✓	87.05 / 8.56	88.31 / 7.54	86.33 / 8.91	66.00 / 26.73
I. Shared Encoder-Decoder	✗	—	—	84.12 / 11.24	85.33 / 10.17	83.75 / 12.05	64.20 /27.71
J. w/o SANS (Sec. 3.3)	✓	Top-2	✓	86.92 / 8.71	89.02 / 7.21	86.34 / 8.69	65.40 /26.33

We extend the ablation to Lung (CT) with the training/inference protocol unchanged. The Full OAMoE baseline (A) attains Brain 88.51 / 7.53, Breast 89.93 / 6.44, Ovary 87.91 / 7.84, Lung 67.60 / 23.52 (Dice / HD95). Replacing anatomy-aware pools with Random Pool (B) or a Single Generic Pool (C) hurts all organs, most notably Lung: 65.70 / 25.92 (ΔDice = −1.90, ΔHD95 = +2.40 vs A) and 65.55 / 26.26 (−2.05, +2.74). A Global Pool (H) also degrades Lung (66.00 / 26.73; −1.60, +3.21), and collapsing to a Shared Encoder–Decoder (I) is worse (64.20 / 27.71; −3.40, +4.19). These results show that anatomical priors are especially important when CT style and lung lesion morphology differ from MRI domains.

Average Fusion (D) underperforms on every organ (e.g., Brain 87.21 / 8.34, Lung 66.20 / 25.45). Top-1 only (F) also trails (Lung 67.20 / 24.88). Thus, two complementary experts offer better accuracy and boundaries. Fixed Top-2 (E) is competitive on Lung (67.60 / 24.71; ΔDice = 0.00, ΔHD95 = +1.19 vs A) and suits its concentrated phenotype distribution. Dynamic Top-2 remains best for Brain (88.51 / 7.53 vs 87.65 / 8.05 for Fixed), reflecting higher sequence heterogeneity. Removing sparsity (G) yields consistent declines (e.g., Lung 66.90 / 25.13; −0.70 Dice, +1.61 HD95), indicating that sparse gating prevents diffuse activations.

Disabling SANS (J) reduces robustness across domains: Brain 86.92 / 8.71 (−1.59, +1.18), Breast 89.02 / 7.21 (−0.91, +0.77), Ovary 86.34 / 8.69 (−1.57, +0.85), Lung 65.40 / 26.33 (−2.20, +2.81). SANS stabilizes internal statistics and improves generalization. Across four organs, anatomy-aware pooling, attention-guided Top-2 cooperation, sparsity, and SANS drive accuracy and boundary stability. Fixed Top-2 is a strong, efficient choice for Lung CT; Dynamic Top-2 remains preferable for Brain MRI.

Finally, our organ-aware hierarchical design significantly outperformed both a generic MoE with a single global pool and a strong shared encoder–decoder baseline (nnUNet), achieving over +4% Dice and −4mm HD95 improvements on Brain, which highlights that the observed gains stem not from parameter scaling but from the proposed architecture itself. Overall, these results clearly demonstrate that the three components act synergistically to enable OAMoE to achieve state-of-the-art performance in different segmentation tasks.

5. Discussion

Despite the robust performance of OAMoE across datasets, several limitations merit further investigation. Recall is suboptimal on several benchmarks. This suggests a conservative behavior in uncertain regions despite competitive overall accuracy. This conservatism could stem from the

blurring of edges observed in medical datasets, which hampers the model's capacity to accurately delineate boundaries. Incorporating an edge-aware module (e.g., BA-Net [51]) may improve boundary recall.

Additionally, the training costs associated with our model are significant. The training duration for each expert pool approximates 40 hours, which constrains scalability, particularly when incorporating new modalities or anatomical regions. While the specialization of experts enhances task-specific performance, it compromises training efficiency compared to more generalized models like nnUNet or MedSAM, which require only a single network training cycle. Implementing strategies such as expert pruning for knowledge distillation [52] or early stopping mechanisms [53] could potentially reduce these training overheads.

Furthermore, the default image resolution of 512×512 pixels, though empirically selected, may not be universally optimal for all organs or imaging modalities. This resolution strikes a balance between preserving context and managing GPU memory costs. However, prior research suggests that certain diminutive structures, such as micro-lesions in breast or ovarian ultrasound/CT, may benefit from multi-scale training or localized patch refinement [54].

6. Conclusion

To address the generalization challenge of multi-organ/multi-modal segmentation under anatomical and modal differences, we propose OAMoE (anatomical prior + expert pool + Top-2 Asymmetric Attention Routing + SANS). Among them, the expert pool of anatomical perception reduces cross-modal interference. Top-2 collaboration is superior to single expert/average fusion, and the division of expertise of experts is visually displayed. Evaluated on 3 organs and 7 datasets. These experiments validate the model's capacity to generalize to unseen datasets and organs, sustaining competitive Dice and low boundary error under cross-domain shifts. Future work will introduce boundary/edge modules, distillation/pruning for cost reduction, multi-scale or local refinement, and expand the expert pool and interpretability to support clinical implementation.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: